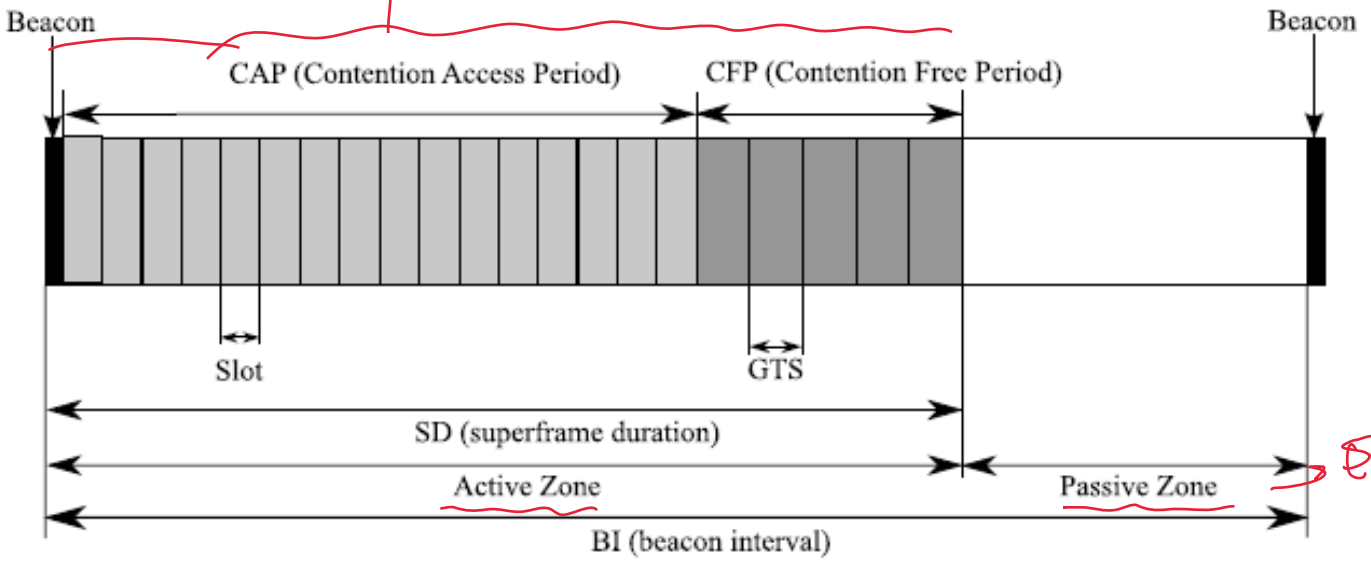
Becon-Enabled Mode

- The use of beacon packets allows to synchronize tne nodes.
- The time is divided into superframes.
- The nodes have, therefore, higher complexity, but duty cycle and channel access are better managed (GTSs).



Beaconless Mode





- No synchronization between nodes.
- Nodes' structures are simpler, but, obviously the performance of beacon-enabled networks is better.
- The MAC protocol differs in terms of time management, synchronization, and packet retransmissions.

CSMA/CA MAC Protocol: Unslotted Version

- The source node waits for a number of slots randomly generated in the interval $[0, 2^{BE_i} - 1]$.
- Once elapsed this time, the node verifies the channel status through a Clear Channel Assessment (CCA), i.e., the energy on the RF interface is measured over $8T_{sym}$.
- If the energy is above a threshold, the channel is busy; otherwise, it is assumed idle.
- If channel is idle, the node can transmit a packet in its queue. Otherwise, NB (number of backoff cycles for this packet) and m (maximum number of backoff cycles, denoted as backoff stage) are compared.
- If $NB = m$, the CSMA/CA is declared as failed. Otherwise, BE is incremented by 1. If $BE_i < BE_{max}$, the procedure is repeated.

CSMA/CA MAC Protocol: Slotted Version



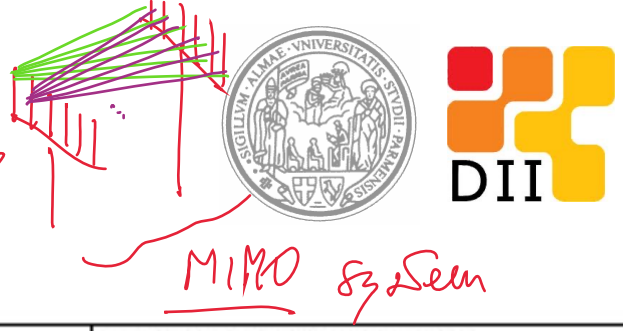


- The main difference is that all operations must be performed in the correspondence of a backoff boundary.
- CCAs of different nodes can collide. Therefore, two CCAs should be performed to prevent from collisions. However, this leads to waste in channel utilization.
- Obviously, in all cases collisions are still possible!!!

IEEE 802.11: Overview

- IEEE 802.11: wireless communication standard originally designed (late 90's) as a high-speed, short-range communication mechanism for laptops and general purpose PCs.
- WiFi is a brand name of the WiFi Alliance: identify equipment and software that is compatible with other WiFi and 802.11 systems
- Number of 802.11 devices worldwide: billions
- The widespread adoption of 802.11 makes deployment of smart objects easy
- Availability of 802.11 chipsets, routers, and network access cards reduces the cost of hardware for 802.11-enabled smart objects.
- Power consumption: typically one order of magnitude higher than that of 802.15.4 devices → new generation of low-power 802.11 transceivers (significant improvement of power consumption during sleeping periods and low power mode: turn off most of the functions, but resume quickly)

IEEE 802.11: standards



IEEE 802.11	Release Date	Frequency (GHz)	Max Data Rate (Mbps)	Range (m)		Status and Comments
				Indoor	Outdoor	
-1997	1997	2.4	2	20	100	Obsolete
a	1999	5/3.7	54	35/-	120/5k	Legacy systems
b	1999	2.4	11	35	140	Legacy systems
g	2003	2.4	54	38	140	Legacy systems
n	2009	2.4/5	600	70	250	Current systems
ac	2013	2.4/5	450/7,000	35		Next generation – just starting to be deployed.
ad	2012	60	7,000	10		Known as WiGig. Short-reach high data rate data transfers.
af	Est. 2016	0.470-0.710	568		6,000	Being called White-Fi because it uses unused TV spectrum.
ah		0.9	40			In development.
aj		45/60	7,000	10		Modification of 802.11ad for 45GHz band for use in China
ax	Est. 2019	2.4/5	450/10,000	35		Revision to 802.11ac to increase efficiency.

IEEE 802.11: foci of the standards

Task Group	Purpose	Status
PHY	Develop Three Physical Layer Options at 1–2 Mbps	Standard
MAC	Develop One Common MAC Layer	Standard
802.11a	High-Speed (54 Mbps) Physical Layer Extension in the 5.8 GHz Band	Standard
802.11b	High-Speed (11 Mbps) Physical Layer in the 2.4 GHz Band	Standard
802.11e	MAC Enhancements for QoS	Standard
802.11g	Even Higher Speed (54 Mbps) Physical Layer Extension in the 2.4 GHz Band	Standard
802.11i	MAC Enhancements for Enhanced Security	Standard
802.11n	Very High-Speed (100 Mbps) Physical Layer	Draft
802.11r	Mobility Handoffs	Draft
802.11s	Mesh Networks	Draft

IEEE 802.11: Low-Power

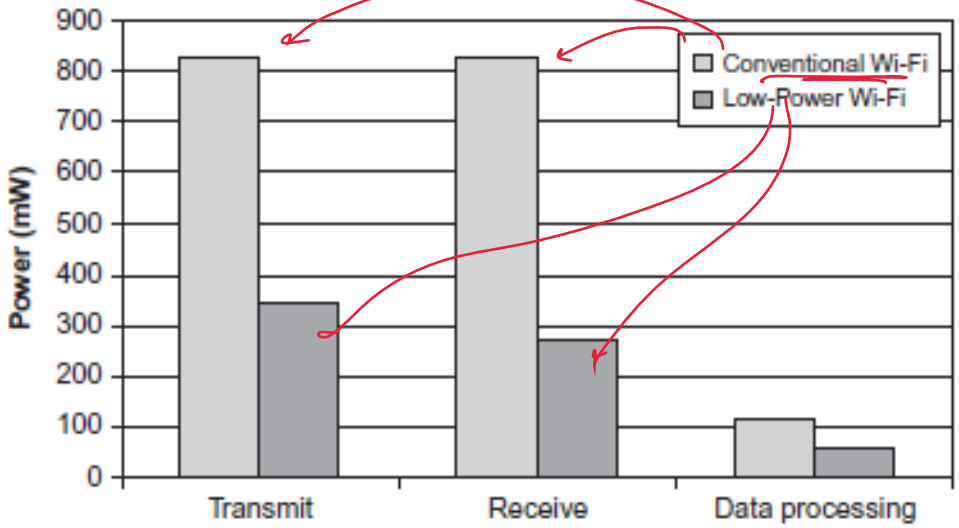


FIGURE 12.15

Comparison of power consumption in transmit, receive, and processing mode for conventional 802.11 and low-power 802.11.

Table 12.1 Power Consumption of Standby and Sleep Modes of a Conventional 802.11 Transceiver and of a Low-power 802.11 Transceiver

	Conventional 802.11 (mW)	Low-power 802.11 (mW)
Standby mode	N/A	0.018
Sleep mode	13	0.2

Note: Conventional 802.11 transceivers do not have a standby mode.

IEEE 802.11: network topology

- Two operational modes: infrastructure (access points) and ad-hoc (direct communications)

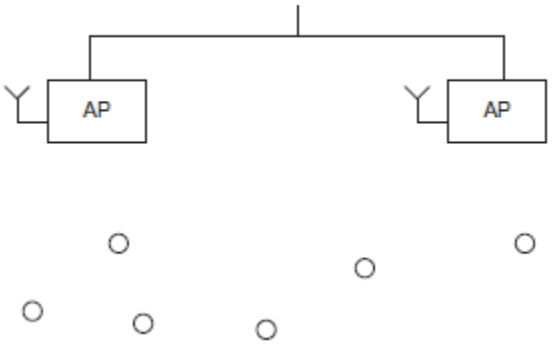


FIGURE 12.11
A network topology of 802.11 in access point mode. Each 802.11 transceiver is associated with an access point. A network may be served by more than one access point. The access points typically have wired network access.

- A node must associate itself with the AP → a group of nodes with an AP is called Basic Service Set (BSS) → with more than one AP, Extended Service Set (ESS) → Service Set IDentifier (SSID)
  - Node probes for a known or unknown SSID
  - AP replies with a probe response packet, with configuration parameters
  - If the network is ok, the node sends an authentication request frame
  - AP replies with association reply frame

IEEE 802.11: PHY layer

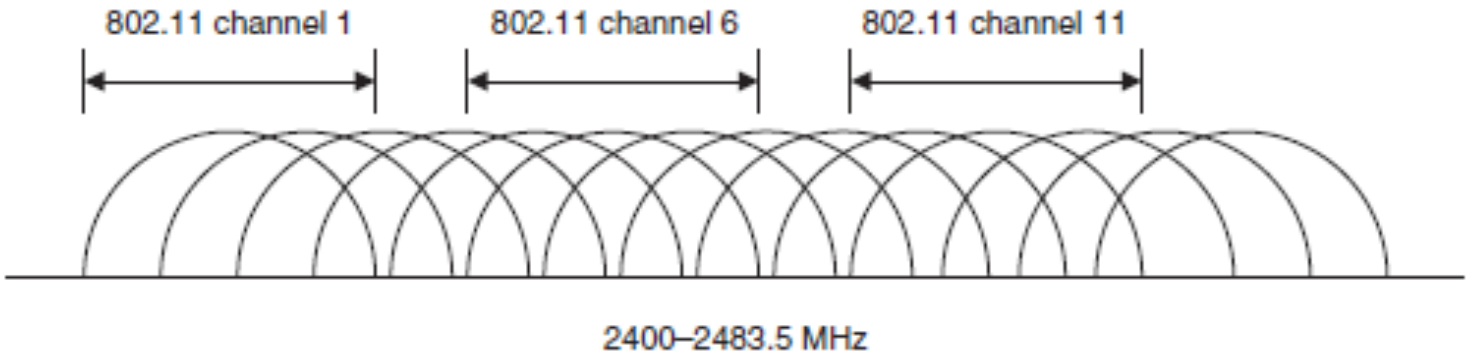


FIGURE 12.12

IEEE 802.11 operates on 14 different channels in the 2.4 GHz band. Channels 1, 6, and 11 do not overlap. Channels 12, 13, and 14 are not available in every country due to radio frequency licensing restrictions.

- Each 802.11 packet is sent with a fixed bit rate, but the bit rate may differ between packets (higher rate with a good communication channel)
- Broadcast packets are sent with a lower bit rate than packets addressed to a specific host to be compatible with older versions

IEEE 802.11: MAC layer

- MAC layer of 802.11: based on Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

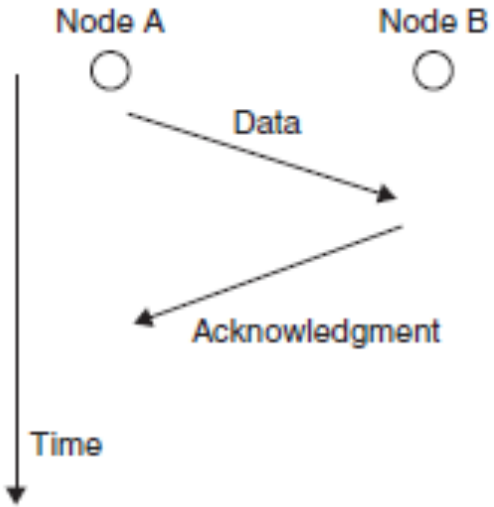


FIGURE 12.13

The IEEE 802.11 MAC layer uses acknowledgment packets that inform the sender that a packet was successfully received by the receiver.

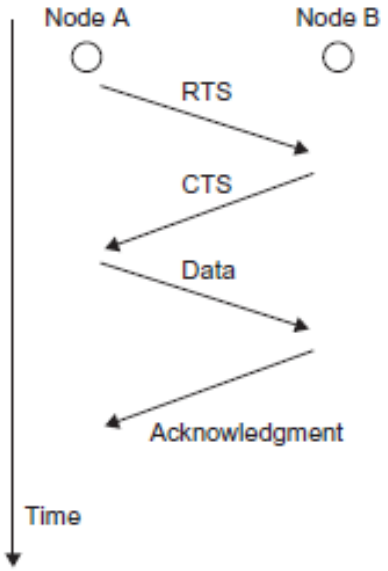


FIGURE 12.14

The RTS/CTS mechanism of the 802.11 MAC layer. To send a packet, a node first sends an RTS packet. If the access point grants the request, it sends a CTS packet and the node can send its packet.

- Mechanism of ready-to-send (RTS) and clear-to-send (CTS) solves the hidden and exposed terminal problems (see later)

PLC: a Quick Overview (1/2)

- Power line communications: eletrical lines are everywhere
- PLC: provides smart objects with data and energy (no need for batteries)

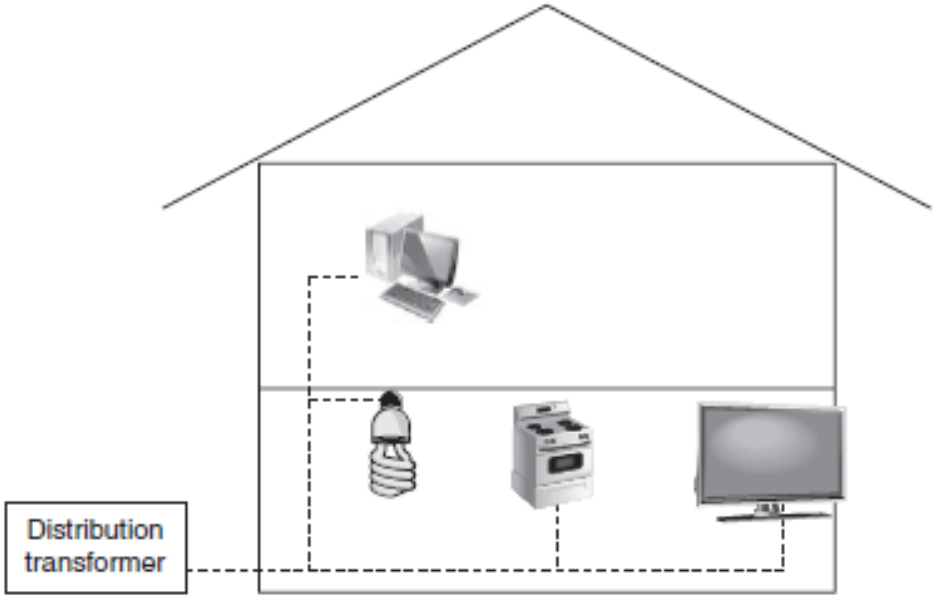


FIGURE 12.16

PLC network within a home.

PLC: a Quick Overview (2/2)



- Data transported across the 50 Hz or 60 Hz electrical distribution network: electrical equipment may add a significant noise
- PLC for short-range communication: only within one electrical domain such as a home or an office building (modulated data signal cannot traverse the electrical voltage transformers outside the electrical domain)
- PHY: copper wires (carry high-power AC signal → noisy; add interferences from plugged-in devices) → very robust communication mechanisms → data rates from kb/s to Mb/s
- MAC: CSMA/CA (unlike transceivers for Ethernet cables, a node transmitting over an electrical line cannot receive)
- Power savings still important: (i) reduce overheating and (ii) avoid compromising overall power savings

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IoT Lab

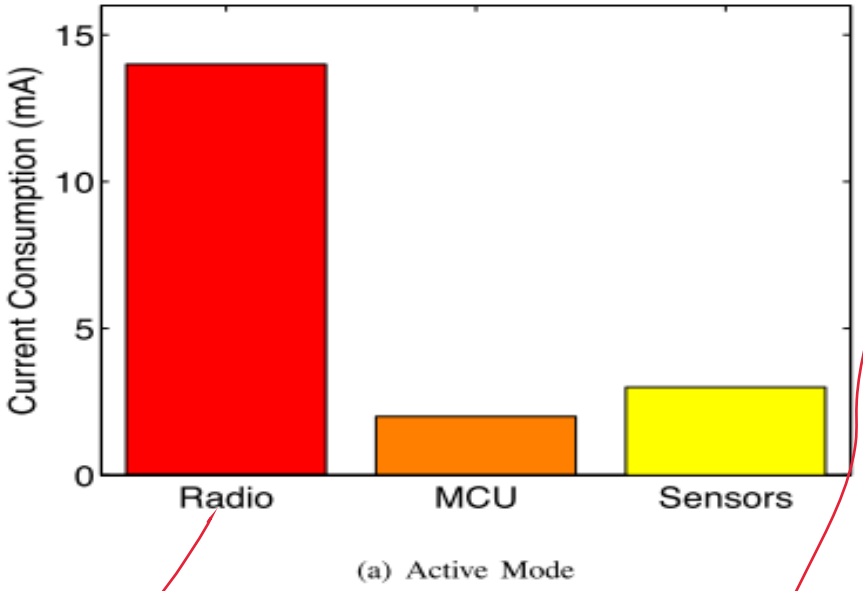
Energy-efficient MAC Protocols

Main sources

- A. Bachir, M. Dohler, T. Watteyne, K. K. Leung, "MAC Essentials for Wireless Sensor Networks," *IEEE Communications Surveys & Tutorials,* vol.12, no.2, pp.222-248, Second Quarter 2010. doi: 10.1109/SURV.2010.020510.00058
- [MACA] P. Karn,"MACA - a New Channel Access Method for Packet Radio", in ARRL/CRRL Amateur Radio 9tt,~ Computer Networking Conference, pp. 134-140, 1990.
- [PAMAS] Suresh Singh and C. S. Raghavendra. 1998. PAMAS—power aware multi-access protocol with signalling for ad hoc networks. *SIGCOMM Comput. Commun. Rev.* 28, 3 (July 1998), 5-26. DOI=10.1145/293927.293928 http://doi.acm.org/10.1145/293927.293928
- [STEM] C. Schurgers, V. Tsiatsis, S. Ganeriwal, M. Srivastava, "Optimizing sensor networks in the energy-latency-density design space," *IEEE Transactions on Mobile Computing*, vol.1, no.1, pp.70,80, Jan-Mar 2002. doi: 10.1109/TMC.2002.1011060
- [B-MAC] J. Polastre, J. Hill, and D. Culler, "Versatile low power media access for wireless sensor networks." Proc. 2nd international conference on Embedded networked sensor systems (SenSys '04), pp. 95-107, Baltimore, MA, USA November 2014. doi>10.1145/1031495.1031508
- [S-MAC] W. Ye, J. Heidemann, D. Estrin, "An energy-efficient MAC protocol for wireless sensor networks," *Proc. 21st Annual Joint Conference of the IEEE Computer and Communications Societies* (INFOCOM 2002) , pp.1567-1576, 2002. doi: 10.1109/INFCOM.2002.1019408
- [X-MAC] Michael Buettner, Gary V. Yee, Eric Anderson, and Richard Han. 2006. X-MAC: a short preamble MAC protocol for duty-cycled wireless sensor networks. In *Proceedings of the 4th international conference on Embedded networked sensor systems* (SenSys '06). ACM, New York, NY, USA, 307-320. DOI=10.1145/1182807.1182838
- [ContikiMAC] A. Dunkels, "The ContikiMAC Radio Duty Cycling Protocol", SICS Technical Report T2011:13. ISSN 1100-3154. December 2011

Power consumption

- Power consumption is one of the biggest challenges because:
  - nodes typically have a limited source of energy (finite battery);
  - battery difficult to replace (and to waste).



(a) Active Mode

Radio mode	Power consumption (mW)
Transmit (T_x)	14.88
Receive (R_x)	12.50
Idle	12.36
Sleep	0.016

3 orders of magnitude reduction

WeMote — Transmission, reception, idle listening

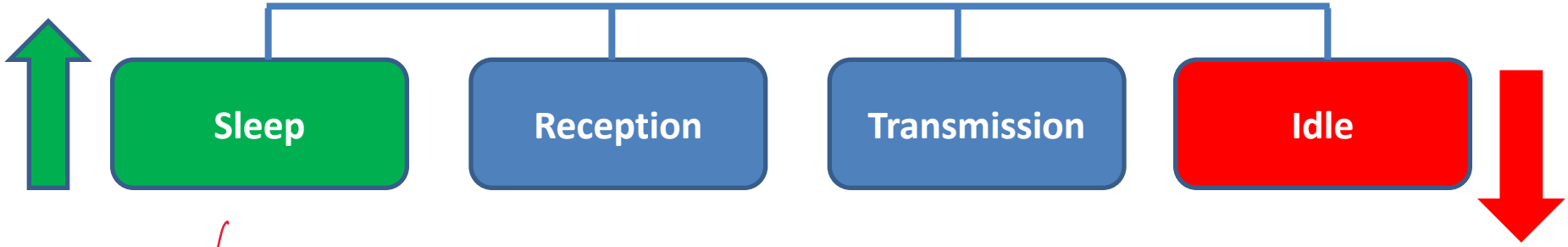
Sources of power consumption

- Useful power consumption:
  - Transmitting or receiving data.
  - Processing queries requests.
  - Forwarding queries and data to the neighbours.
- Wasteful power consumption:
  - Idle listening to the channel "waiting for possible traffic".
  - Retransmitting because of collisions
  - Overhearing: a node receives (and processes) a packet which was not destined to it.
  - Generating and handling control packets (overhead).

Power consumption in WSNs

How to minimize the energy consumption of sensor nodes while meeting the application requirements?

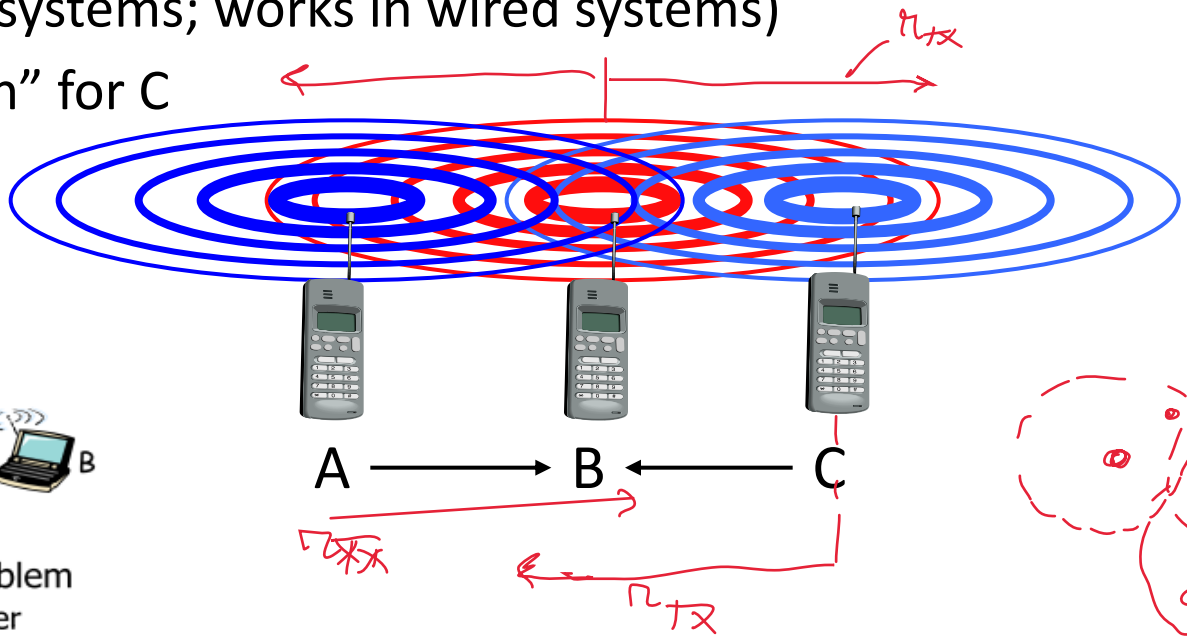
- Use Protocols that aim mainly to increase the sleep periods as much as possible

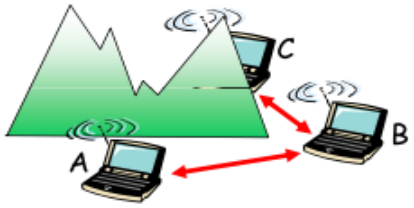


↳ The typical sleeping portion considered for IoT systems may go up to 99.9%

Hidden Terminal Problem

- A sends to B, C cannot receive A
- C wants to send to B, C senses a “free” medium (CS=carrier sensing fails)
- collision at B, A cannot receive the collision (CD=collision deteciont fails in wireless systems; works in wired systems)
- A is “hidden” for C





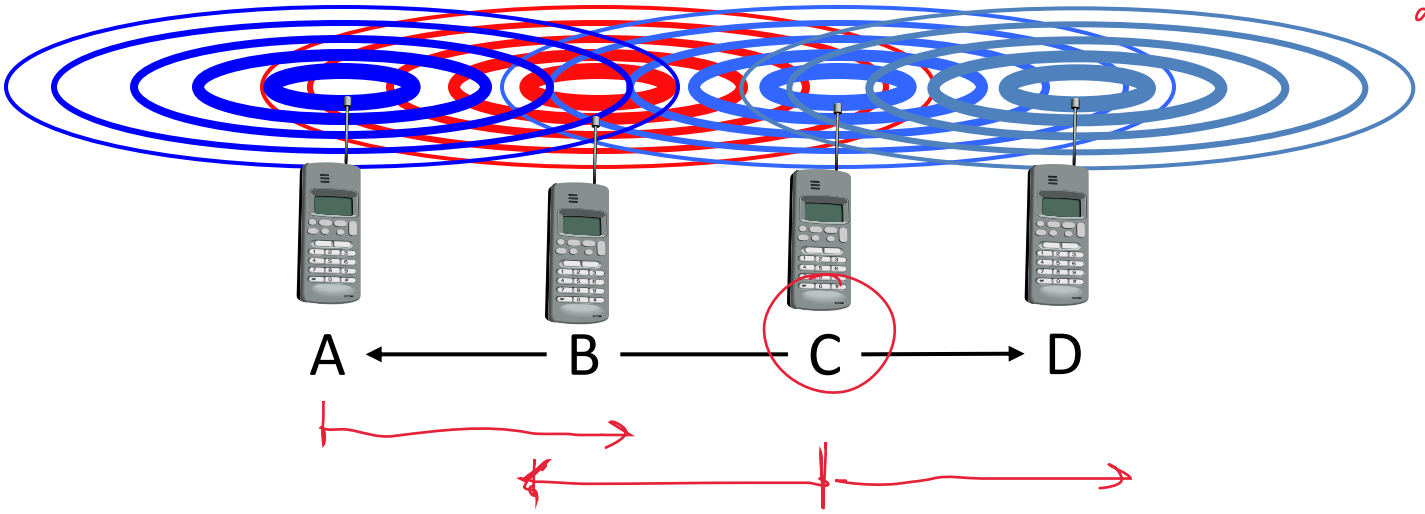
Hidden terminal problem
- B, A hear each other
- B, C hear each other
- A, C can not hear each other

means A, C unaware of their interference at B



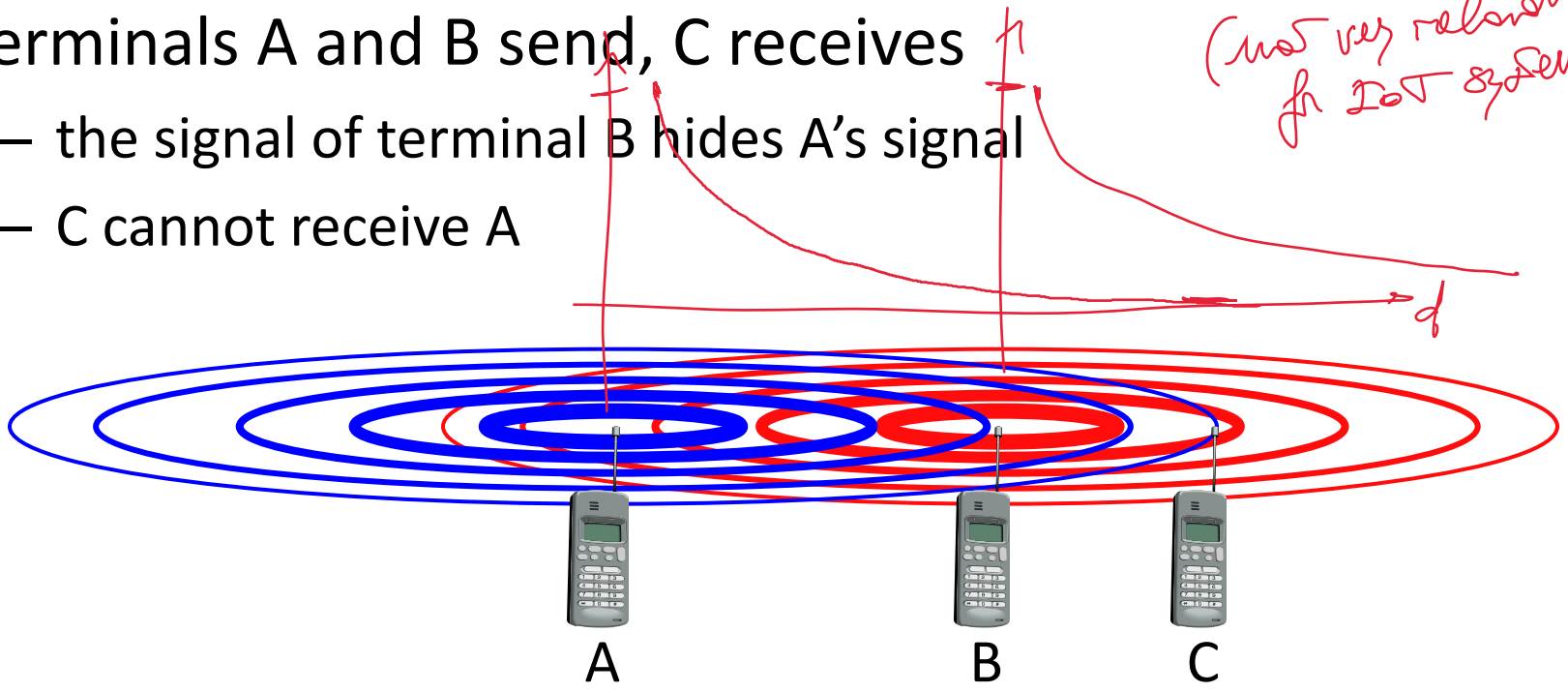
Exposed Terminal Problem

- B sends to A, C wants to send to D
- C has to wait, as it senses a medium in use
- since A is outside the radio range of C waiting is not necessary
- C is "exposed" to B → C is prevented from transmitting even if its transmission would not damage B→A.



Near-Far Problem

- Terminals A and B send, C receives
  - the signal of terminal B hides A's signal
  - C cannot receive A

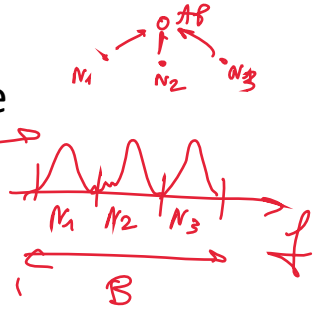


- This is also a severe problem for CDMA networks
- Precise power control required

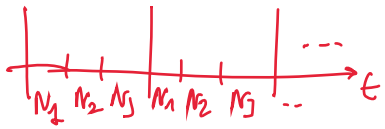
Centralized Access Methods (Cellular)



- SDMA (Space Division Multiple Access)
  - Segment space into sectors, use directional antennas
  - Use cells to reuse frequencies
- FDMA (Frequency Division Multiple Access)
  - Assign a certain frequency to a transmission channel for the entire transmission duration
  - Permanent (radio broadcast), slow hopping (GSM), fast hopping (FHSS, Frequency Hopping Spread Spectrum → bluetooth)



- TDMA (Time Division Multiple Access)
  - Assign a fixed sending frequency (possibly the entire spectrum) for a certain amount of time
- CDMA (Code Division Multiple Access)
- Combinations!

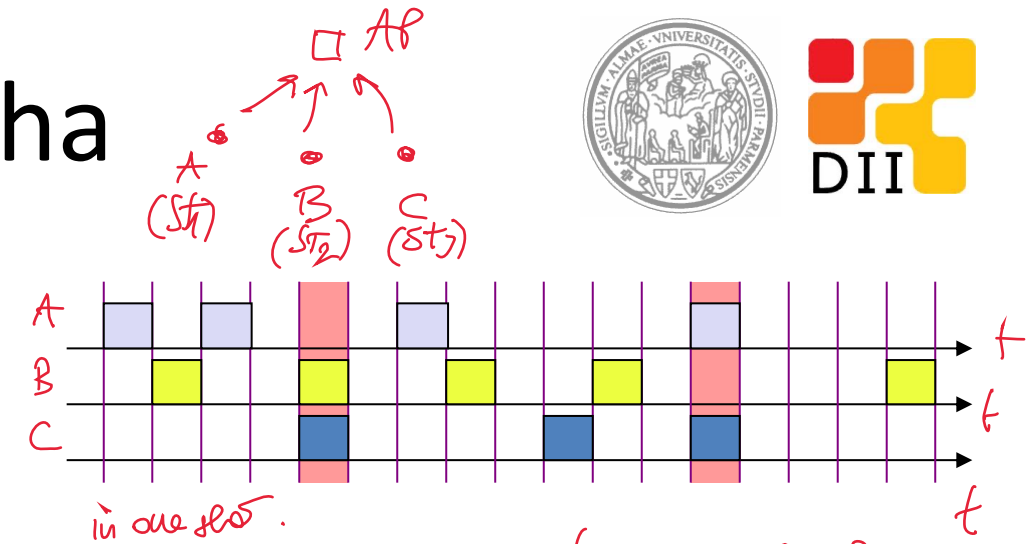




Traditional MAC Protocol Classification

- Centralized/Single-Hop Protocols (see previous page)
  - A base station/controller coordinates all traffic (SDMA, FDMA, TDMA, CDMA)
- Contention Protocols (Aloha, Carrier Sense Multiple Access-CSMA)
  - Transmit when you feel like transmitting (Aloha)
  - Retry if collision, try to minimize collisions, additional reservation modes (CSMA)
  - Problem: Receiver must be awake as well
- Scheduling Protocols
  - Use a "pre-computed" schedule to transmit messages
  - Distributed (non-centralized) and adaptive solutions are difficult
- Hybrid protocols
  - E.g. contention with reservation → scheduling (e.g., IEEE 802.15.4 with beacons)
  - Specific ("cross-layer") solutions

Slotted Aloha



- We assume that the stations are perfectly synchronous
- In each time slot each station transmits with probability *p*.

All stations are independent

$$P_1 = \Pr[\text{Station 1 succeeds}] = p(1-p)^{n-1}$$

$$P = \Pr[\text{any Station succeeds}] = nP_1$$

$$\text{maximize } P: \frac{dP}{dp} = n(1-p)^{n-2}(1-pn) \overset{!}{=} 0 \Rightarrow pn = 1 \Rightarrow p^* = \frac{1}{n}$$

$$\text{then, } P = (1-\frac{1}{n})^{n-1} \geq \frac{1}{e}$$

$P = n \cdot p \cdot (1-p)^{n-1}$, $p \in (0,1)$

$P(p^*) = \binom{n}{1} \frac{1}{n}\left(1-\frac{1}{n}\right)^{n-1}$

$P\{\{St_1 \text{ succeeds}\} \cup \{St_2 \text{ succeeds}\} \cdots \cup \{St_n \text{ succeeds}\}\} = n \cdot P_1$

disjoint events

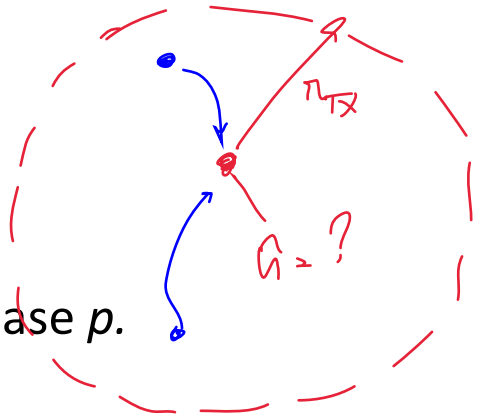
- In Slotted Aloha, a station can transmit successfully with probability at least 1/*e*, i.e., about 36% of the time.
- In pure Aloha (unslotted), a station can transmit successfully with probability at least 1/2e, i.e., about 18% of the time.

$\left(1-\frac{1}{n}\right)^{n-1} \xrightarrow{n\to\infty} e^{n\cdot(-\frac{1}{n})} = e^{-1}$

$e^{\ln(1-\frac{1}{n})^{(n-1)}} = e^{(n-1)\ln(1-\frac{1}{n})}$, $\ln(1-x) \xrightarrow[x\to 0]{} -x$

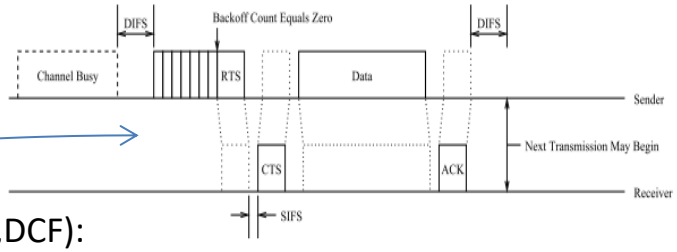
Adaptive Slotted Aloha

- Idea: change the access probability as a function of the number of stations
- How can we estimate the current number of stations in the system?
- Assume that stations can distinguish whether 0, 1, or more than 1 stations transmit in a time slot.
- Model:
  - Number of stations that want to transmit: n.
  - Estimate of n: $\hat{n}$
  - Transmission probability: $p = 1/\hat{n}$
- Idea:
  - If you see that nobody transmits, increase p.
  - If you see that more than one transmits, decrease p.



Carrier Sense Multiple Access (CSMA)

- Listen before transmit: if the channel is free, then transmit
- Various versions:
  - non-persistent: if the device detects activity on the channel, it performs a back off by waiting before attempting to transmit.
  - *p*- persistent: if it detects activity on the channel, it keeps on sensing the channel instead of delaying.
- "Basic" CSMA requires devices to remain in the receive state (not idle) when not transmitting → the transceiver consumes energy too quickly.
- Various variations
  - CSMA/CD: collision detection (Ethernet, IEEE 802.3)
  - CSMA/CA: collision avoidance, LIFO (IEEE 802.15.4, IEEE 802.11-PCF,DCF): combine with Ready-to-send (RTS) and Clear-to-Send (CTS) mechanisms → RTS messages may collide
  - IEEE 802.11 uses Network Allocation Vector (NAV), virtual carrier-sensing mechanism: the MAC frame header contains the packet transmission duration



Backoff

- Backoff protocols rely on acknowledgements.
- Binary exponential backoff:
  - If a packet has collided k times (no ACK for k times), we set $p = 2^{-k}$
  - Or alternatively: wait from random number of slots in $[1,...,2^k]$
- It has been shown that binary exponential backoff is not stable for any arrival rate $\lambda > 0$ if there are infinitely many potential stations
- Interestingly when there is a finite number of stations, binary exponential backoff becomes unstable with $\lambda > 0.568$;
  Polynomial backoff however, remains stable for any $\lambda < 1$.

Network Performance

Moving towards WSNs

- Traditional MAC protocols
  - High throughput
  - Low latency
  - Fairness
  - Mobility
  - Energy? → neglected

→

- MAC protocols for WSNs:
  - Primary goals
    - Energy conservation
    - Scalability and adaptivity
  - Secondary goals
    - Throughput
    - Latency
    - Fairness

Energy-efficient MAC Protocols

- In WSNs, energy is often more critical than throughput.
  - The radio transceiver should be turned off as much as possible.
- Energy management considerations have a big impact on MAC protocols.
  - Idle listening costs about as much energy as transmitting
- In the following: a few ideas, taken from some known protocols that try to balance throughput and energy consumption:
  - MAC,PAMAS, STEM, B-MAC, X-MAC, WiseMAC, S-MAC, **ContikiMAC**
  - Many of the hundreds of MAC protocols that were proposed have similar ideas…



Unscheduled MAC protocols



- "Evolutions" of CSMA
- Advantages:
  - they can adapt for changes "in the node density, traffic load or the topology" better than scheduled protocols.
  - The nodes do not have to be synchronized together.
- Disadvantages:
  - All nodes need to listen to the channel
- Examples:
  - MACA (1990)
  - PAMAS (1998)
  - STEM (2002)
  - B-MAC (2004), WiseMAC (
  - X-MAC (2006)

MACA (Multiple Access with Collision Avoidance) - 1990

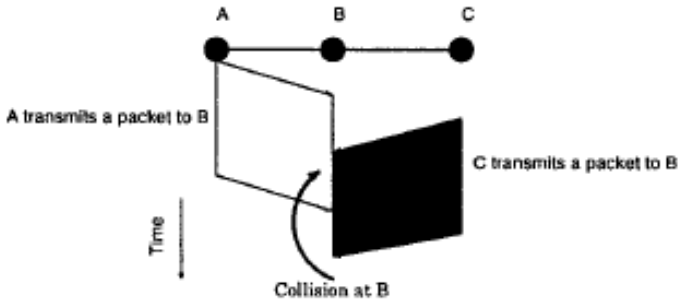
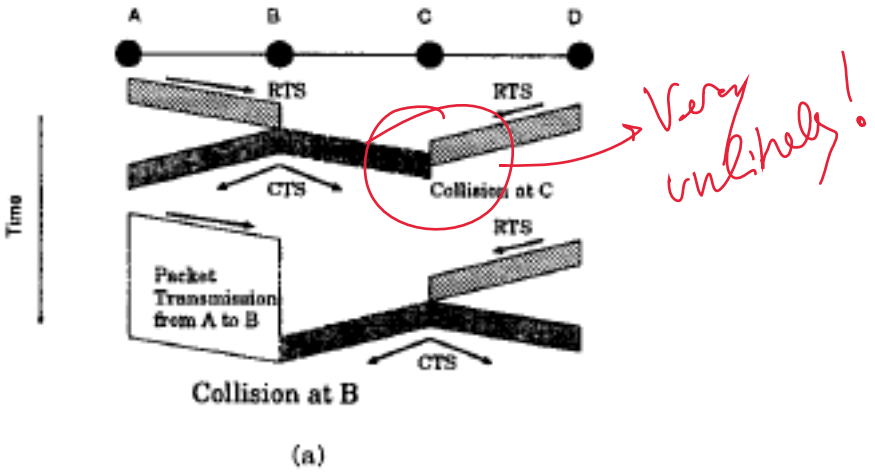


Figure 2: The hidden terminal problem.

- Hidden terminal problem: partially solved with the RTS/CTS mechanism introduced by MACA

Also solves the exposed terminal problem



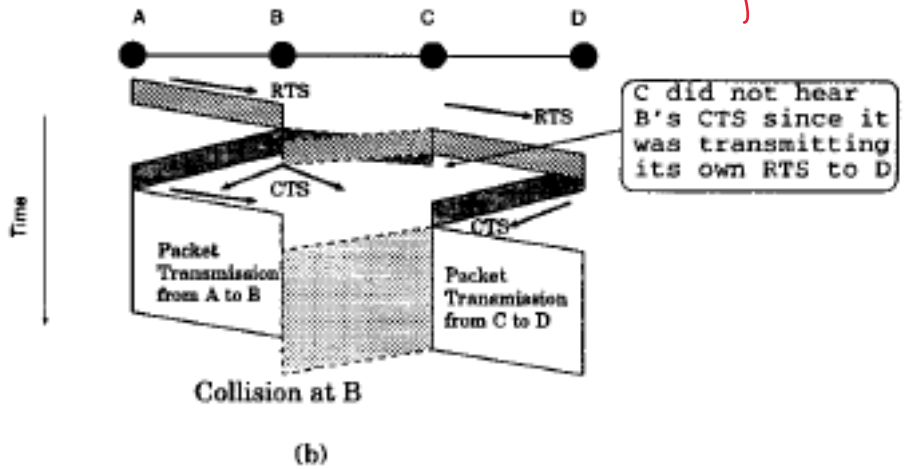


Figure 3: Hidden terminal problem in MACA.

PAMAS (Power Aware Multi-Access protocol with Signaling for Ad-Hoc Networks) - 1998

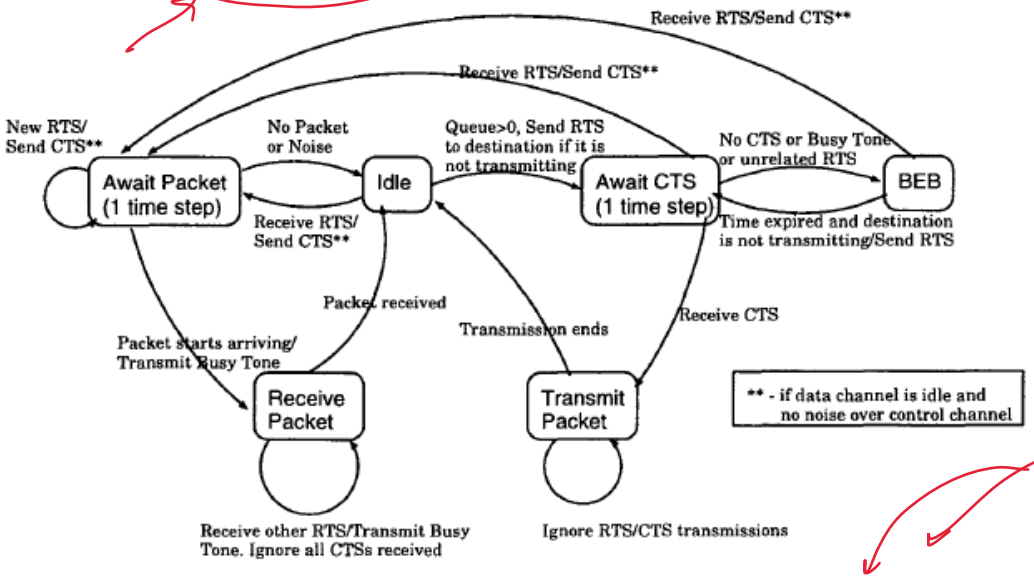
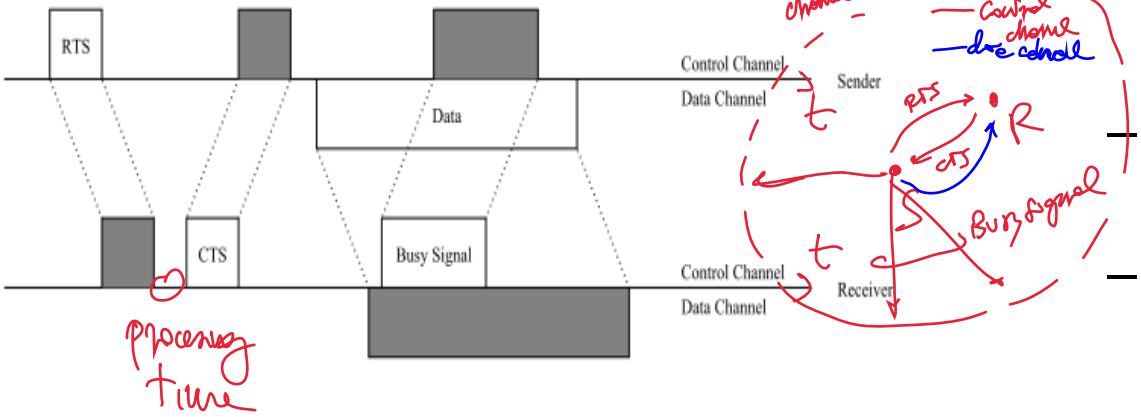


Figure 4: The PAMAS Protocol.



- Basic idea: use a *signaling channel* separated from data channel
- Advantages:
  - Prevent collisions
- Disadvantages:
  - Multiple radio requirement
  - Increase energy consumption
  - Increase device complexity and cost

STEM (Sparse Topology and Energy Management) - 2002

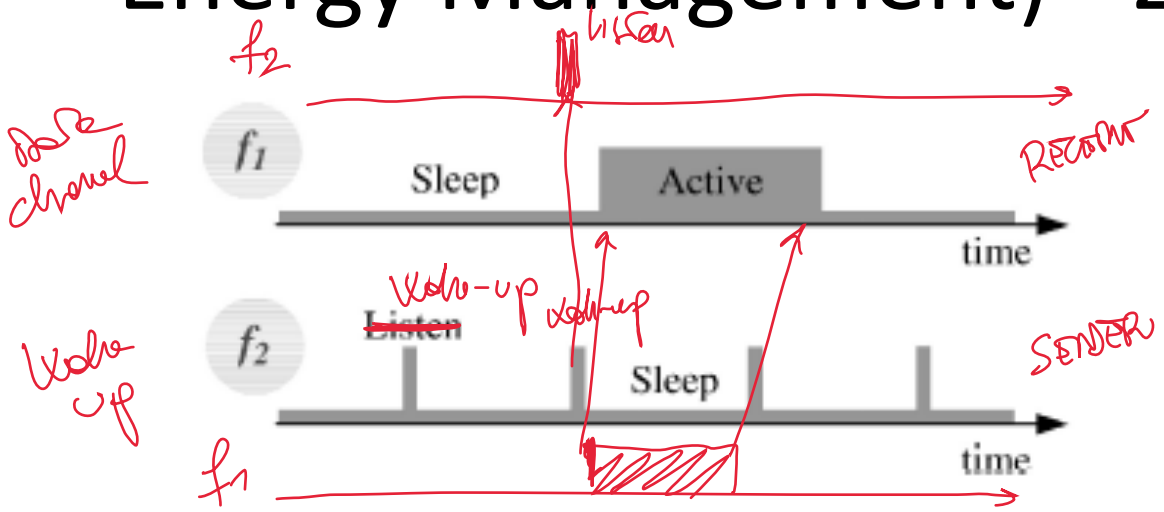


Fig. 3. Separate data and wakeup using two radios.

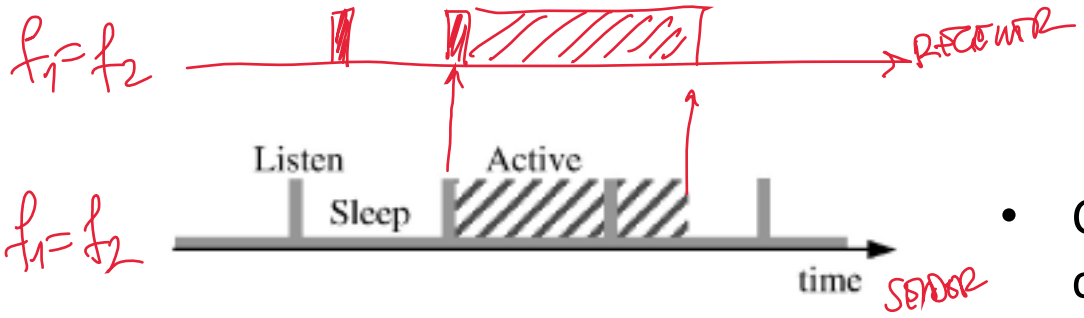


Fig. 4. Separate data and wakeup in time.

- Basic idea: uses two different channels: the wakeup channel (f_2) and the data channel (f_1)
- Requires two transceivers in each node (dual radio)
- Could "mix" the two channels → synchronization is required

STEM-B and STEM-C - 2002

- STEM-B: a sensor node wakes up a neighbor by transmitting a beacon (no RTS/CTS)
  - Advantages:
    - Lower Latency
  - Disadvantages:
    - More complex
    - High energy consumption
- STEM-T: a sensor node wakes a neighbor by transmitting a busy tone (contains no destination address) of sufficient length so that the destination will have a high probability of sensing
  - Disadvantages:
    - High latency
    - Results in overhearing (stops the transmission from nodes which would not interfere)

"Berkeley MAC" (B-MAC) – 2004

(Versatile low power media access for wireless sensor networks)

- Goals
  - Low power operation
  - Effective collision avoidance
  - Simple implementation (small code)
  - Energy-efficient at both low and high data rates
  - Reconfigurable by upper layers
  - Tolerant to changes on the network
  - Scalable to large number of nodes

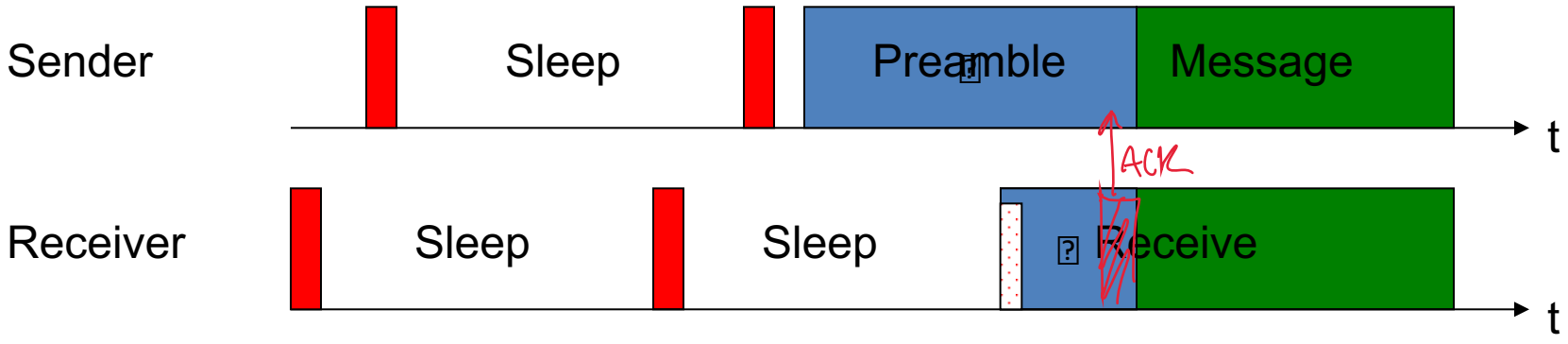
→ MIRACLE !

- Strategy
  - Nodes wake up for a short period and check for channel activity.
  - Return to sleep if no activity is detected. or if they do not need to transmit.
  - If a sender wants to transmit a message, it sends a long preamble to make sure that the receiver is listening for the packet (STEM-T) → preamble has the size of a sleep interval
- Very robust
  - No synchronization required
  - Instant recovery after channel disruption

B-MAC's Features

- Clear Channel Assessment (CCA)
- Low Power Listening (LPL) using preamble sampling
- Hidden terminal and multi-packet mechanisms not provided: they should be implemented, if needed, by higher layers

- CCA on/off
- Acknowledgements on/off
- Initial and congestion backoff on a per-packet basis
- Configurable check interval and preamble length



B-MAC's Lifetime Model

Operation	Time (s)		I (mA)	
Initialize radio (b)	350E-6	$t_{r_{init}}$	6	$c_{r_{init}}$
Turn on radio (c)	1.5E-3	t_{ron}	1	c_{ron}
Switch to RX/TX (d)	250E-6	$t_{rx/tx}$	15	$c_{rx/tx}$
Time to sample radio (e)	350E-6	t_{sr}	15	c_{sr}
Evaluate radio sample (f)	100E-6	t_{ev}	6	c_{ev}
Receive 1 byte	416E-6	t_{rxb}	15	c_{rxb}
Transmit 1 byte	416E-6	t_{txb}	20	c_{txb}
Sample sensors	1.1	t_{data}	20	c_{data}

Table 2: Time and current consumption (I) for completing primitive operations of a monitoring application using the Mica2 mote and CC1000 transceiver. Identifiers on each operation map back to the activities of acquiring a radio sample in Figure 3.

Notation	Parameter	Default
c_{sleep}	Sleep Current (mA)	0.030
C_{batt}	Capacity of battery (mAh)	2500
V	Voltage	3
$L_{preamble}$	Preamble Length (bytes)	271
L_{packet}	Packet Length (bytes)	36
t_i	Radio Sampling Interval (s)	100E-3
n	Neighborhood Size (nodes)	10
r	Sample Rate (packets/s)	1/300
t_l	Expected Lifetime (s)	-

Table 3: Parameters for a monitoring application running B-MAC. The first three parameters are specific to Mica2 motes; the next three are default values for B-MAC parameters on the Mica2; the remaining parameters are application semantics affecting B-MAC's performance. Each parameter affects the node's total energy consumption, E.

$$t_{tx} = r \times (L_{preamble} + L_{packet}) t_{txb}$$
$$E_{tx} = t_{tx} c_{txb} V$$

$$t_d = t_{data} \times r$$
$$E_d = t_d c_{data} V$$

$$t_{sleep} = 1 - t_{rx} - t_{tx} - t_d - t_{listen}$$
$$E_{sleep} = t_{sleep} c_{sleep} V$$

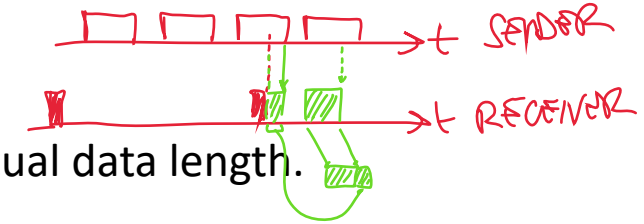
$$E = E_{rx} + E_{tx} + E_{listen} + E_d + E_{sleep}$$

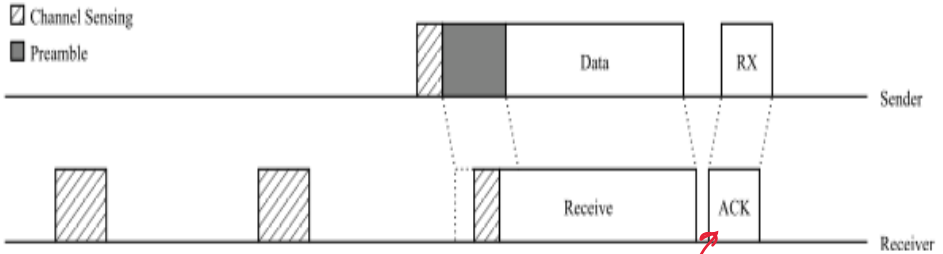
$$t_{rx} \leq nr(L_{preamble} + L_{packet}) t_{rxb}$$
$$E_{rx} = t_{rx} c_{rxb} V$$

$$E_{sample} = 17.3\mu\text{J}$$
$$t_{listen} = (t_{r_{init}} + t_{ron} + t_{rx/tx} + t_{sr}) \times \frac{1}{t_i}$$
$$E_{listen} \leq E_{sample} \times \frac{1}{t_i}$$

B-MAC's Issues

- Problem: All nodes in the vicinity of a sender wake-up and wait for the packet → overhearing
  - Solution 1: Send wake-up packets instead of preamble, wake-up packets tell when data is starting so that receiver can go back to sleep as soon as it received one wake-up packet.
  - Solution 2: Just send data several times such that receiver can tune in at any time and get tail of data first, then head.
- Communication costs are mostly paid by the sender.
  - The preamble length can be much longer than the actual data length.
- Idea: Learn wake-up schedules from neighboring nodes.
  - Start sending preamble just before intended receiver wakes up.
  - **WiseMAC** → encode wake-up pattern in the ACK message





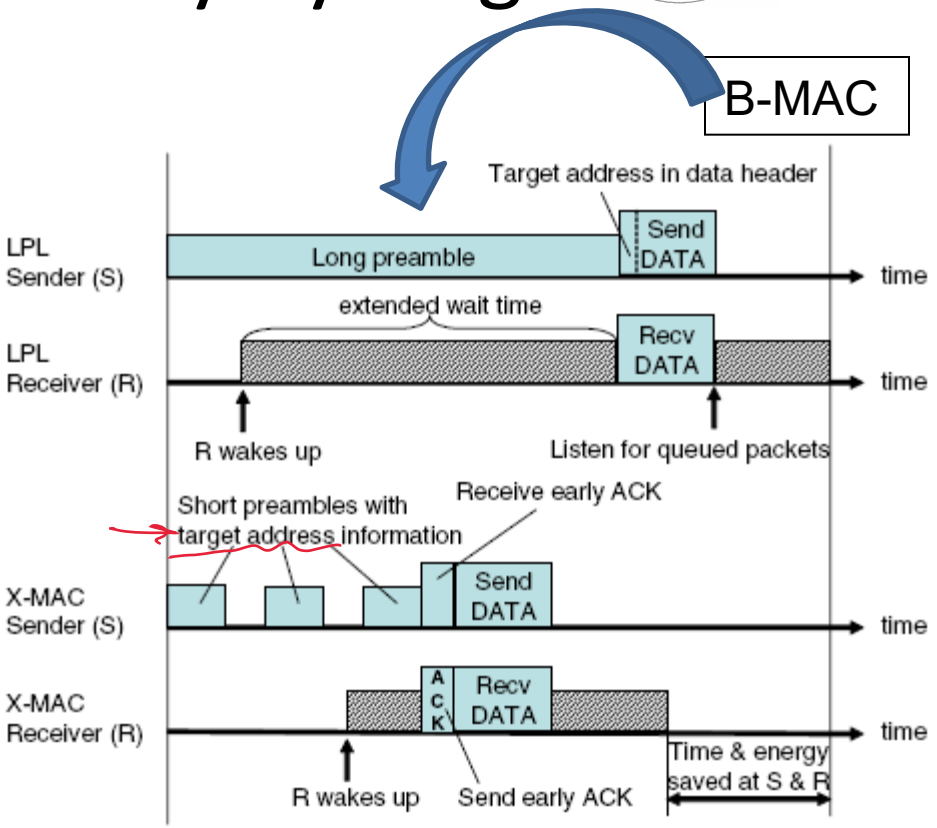
X-MAC: Principles – 2006



- Short preamble
  - Reduce latency and reduce energy consumption
- Target (destination) in preamble
  - Minimize overhearing problem.
- Strobed preamble → the preamble transmission can be interrupted.
  - Reduces latency for the case where destination is awake before preamble completes.
- Reduced per-hop latency and energy
- Dynamic duty-cycling (depending on network conditions)

X-MAC: Asynchronous Duty Cycling

- Overhearing energy loss is proportional to number of receivers in range.
- Short preamble packets.
- Non-receivers return to sleep more quickly.
- **Bad in high density sensor networks** (collisions between strobing packets)



Scheduled MAC protocols



- Idea: try to reduce the energy consumption by *coordinating* sensor nodes with a common schedule
- Advantages:
  - Saving the power from being wasted by turning off the radio outside the allocated time slot.
  - Limits collisions, idle listing, and overhearing
- Disadvantages:
  - When a sensor node enters the network, it must wait till it learns the schedule → some delay exist
  - Cost of increased number of exchanged messages
  - No flexibility to changes in sensor nodes' density or movements.
  - All sensors have to be synchronized.
- Example:
  - S-MAC (2002)
  - ContikiMAC (from 2008 on)

Sensor MAC (S-MAC) - 2002



- Tries to reduce waste of energy from all four sources of energy inefficiency
  1. Collisions – by using RTS and CTS
  2. Overhearing – by switching the radio off when transmission is not meant for that node
  3. Control Overhead – by message passing
  4. Idle listening – by periodic listen and sleep
- At the cost of:
  - some reduction in per-hop fairness and latency
  - but does not reduce end-to-end fairness and latency

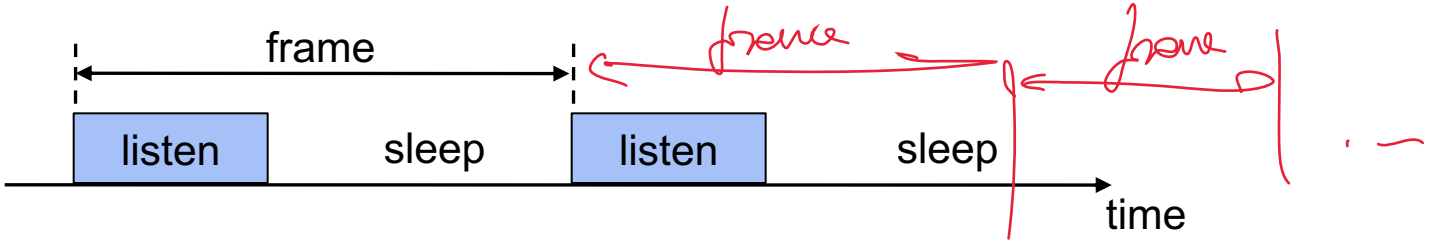
S-MAC: Assumptions

- Network assumptions:
  - Composed of many small nodes deployed in ad hoc fashion (random)
  - Most communications will be between nodes as peers, rather than to/from a single base station
  - Nodes must self-configure → sleep/wake-up schedule
- Application assumptions:
  - Dedicated to a single application or a few collaborative applications
  - Involves in-network processing to reduce traffic and increase life time
  - Applications have long idle periods and can tolerate some latency

S-MAC: Approach

- [Periodic listening and sleeping] Coarse-grained TDMA-like sleep/awake cycles:
  - the sensor node periodically goes to the fixed listen/sleep cycle.
  - A time frame in S-MAC is divided into two parts: one for a listening session and the other for a sleeping session.

increased latency



- [Message passing] All nodes choose and announce awake schedules.
  - synchronize to awake schedules of neighboring nodes.
- [Collision and overhearing avoidance] Uses RTS/CTS to resolve contention during listen intervals.
  - And allows interfering nodes to go to sleep during data exchange.

S-MAC: Possible Scenarios

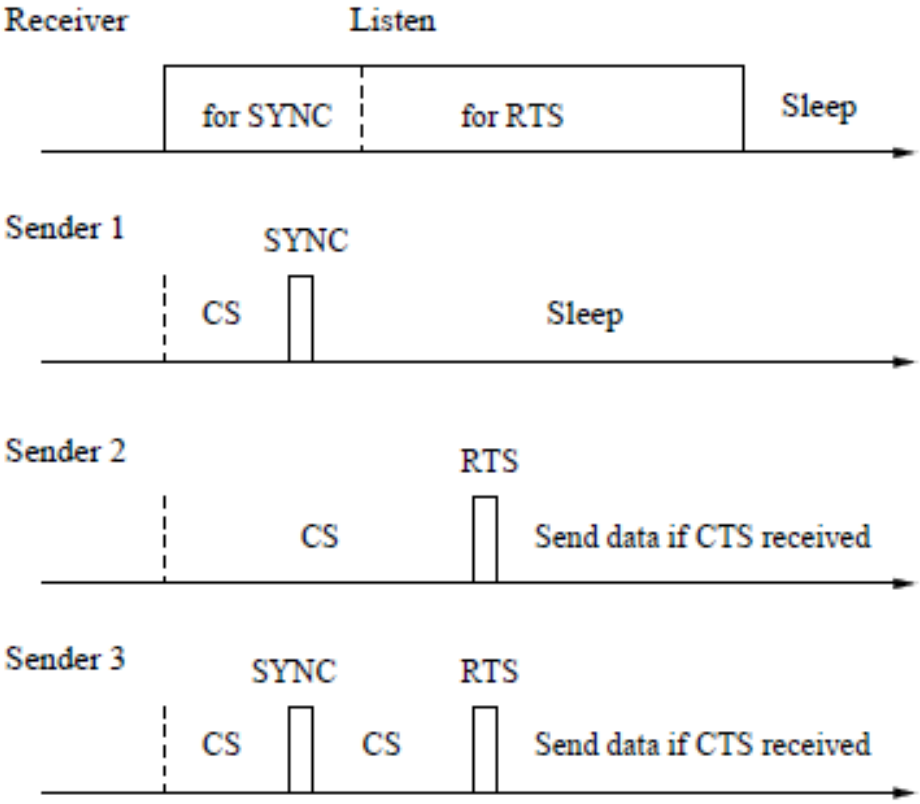
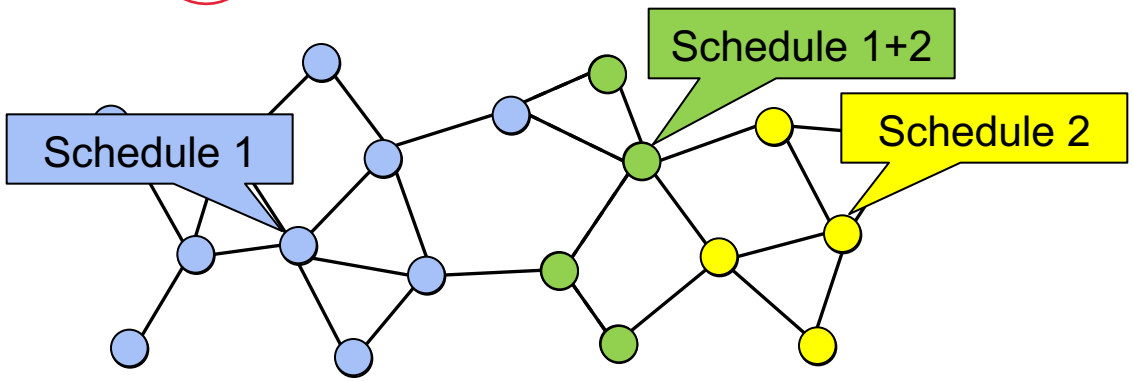


Fig. 3. Timing relationship between a receiver and different senders. CS stands for carrier sense.

Figure 3 also shows the timing relationship of three possible situations that a sender transmits to a receiver. CS stands for carrier sense. In the figure, sender 1 only sends a SYNC packet. Sender 2 only wants to send data. Sender 3 sends a SYNC packet and a RTS packet.

S-MAC: A Problem

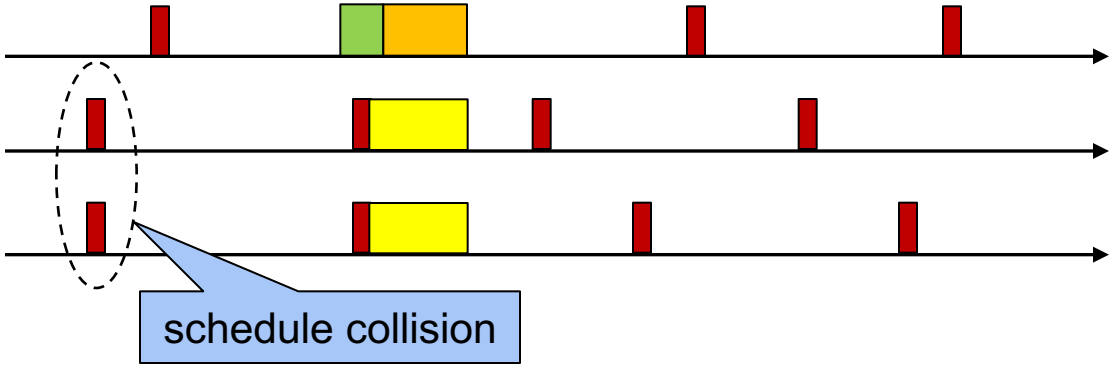
- Nodes may have to follow multiple schedules to avoid network partition.



- A fixed sleep/awake ratio is not always optimal.
  - Variable load in the network.
- Idea: Adapt listen interval depending on the current network load → T-MAC

Hybrid MAC Protocols

- Protocols may use information from upper layers to further improve their performance.
  - Information about neighborhood
  - Routing policies
- Minimize costly overhearing of neighboring nodes
  - Inform them to change their channel sniff patterns



- Use randomization to resolve schedule collisions
- Example: I. Rhee, A. Warrier, M. Aia, and J. Min, "Z-MAC: a Hybrid MAC for Wireless Sensor Networks", Sensys 2005.

ContikiMAC – 2011 (Contiki 2.5)

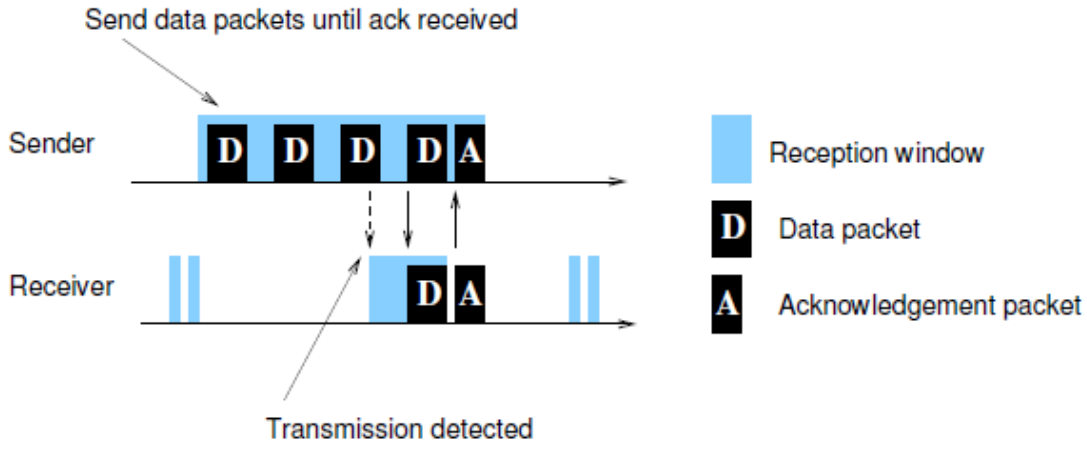


Figure 1: ContikiMAC: nodes sleep most of the time and periodically wake up to check for radio activity. If a packet transmission is detected, the receiver stays awake to receive the next packet and sends a link layer acknowledgment. To send a packet, the sender repeatedly sends the same packet until a link layer acknowledgment is received.

ContikiMAC: Broadcast

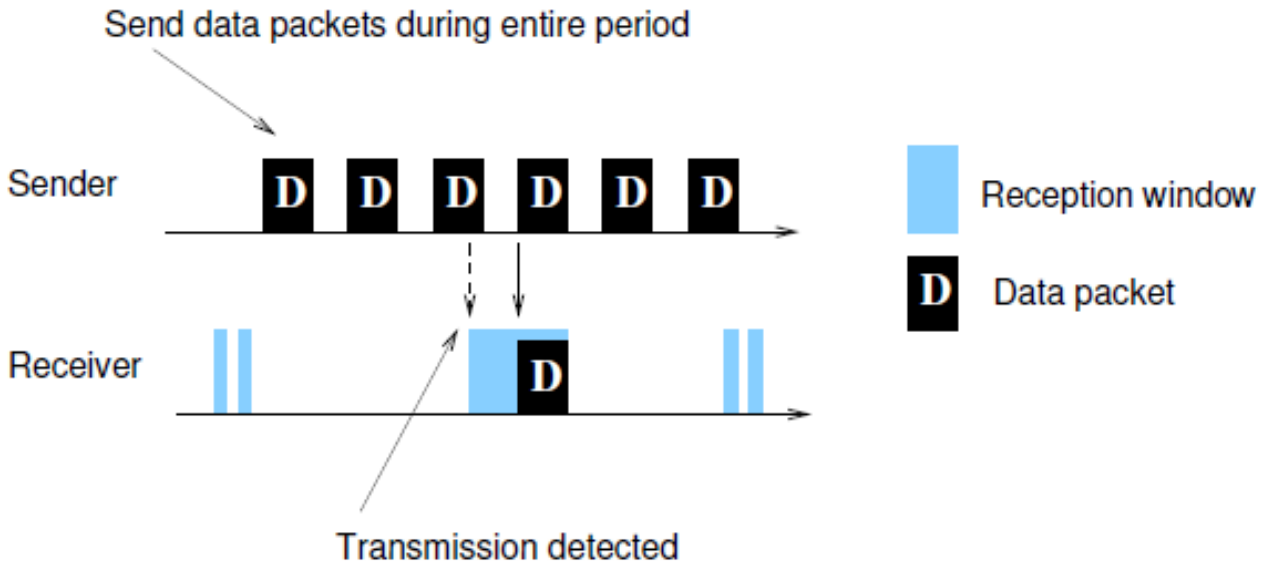


Figure 2: Broadcast transmissions are sent with repeated data packets for the full wake-up interval.

ContikiMAC: CCA timing

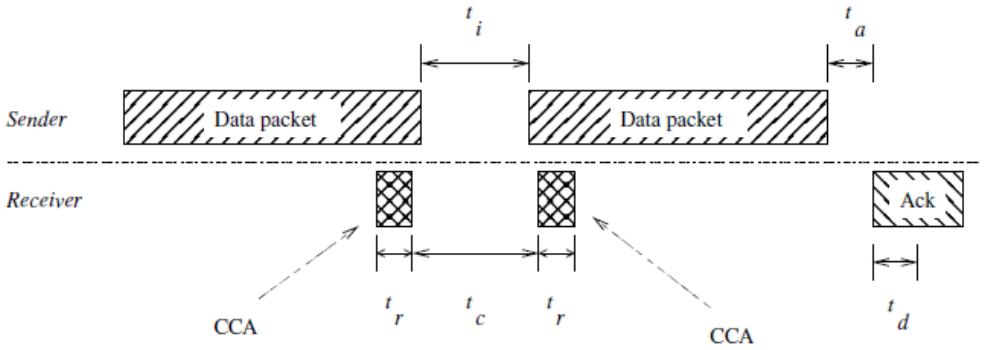


Figure 3: The ContikiMAC transmission and CCA timing.

The ContikiMAC timing is shown in Figure 3. The timing requirements from the figure are:

t_i: the interval between each packet transmission.

t_r: the time required for a stable RSSI, needed for a stable CCA indication.

t_c: the interval between each CCA.

t_a: the time between receiving a packet and sending the acknowledgment packet.

t_d: the time required for successfully detecting an acknowledgment from the receiver.

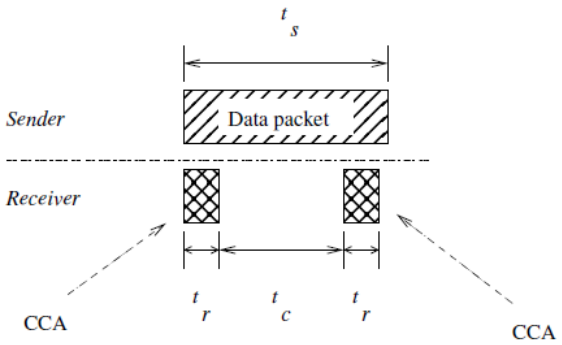


Figure 4: A packet transmission must be long enough so that it does not fall between to subsequent CCAs.

Timing constraints

$$t_a + t_d < t_i < t_c < t_c + 2t_r < t_s.$$

ContikiMAC: Transmission Phase Lock

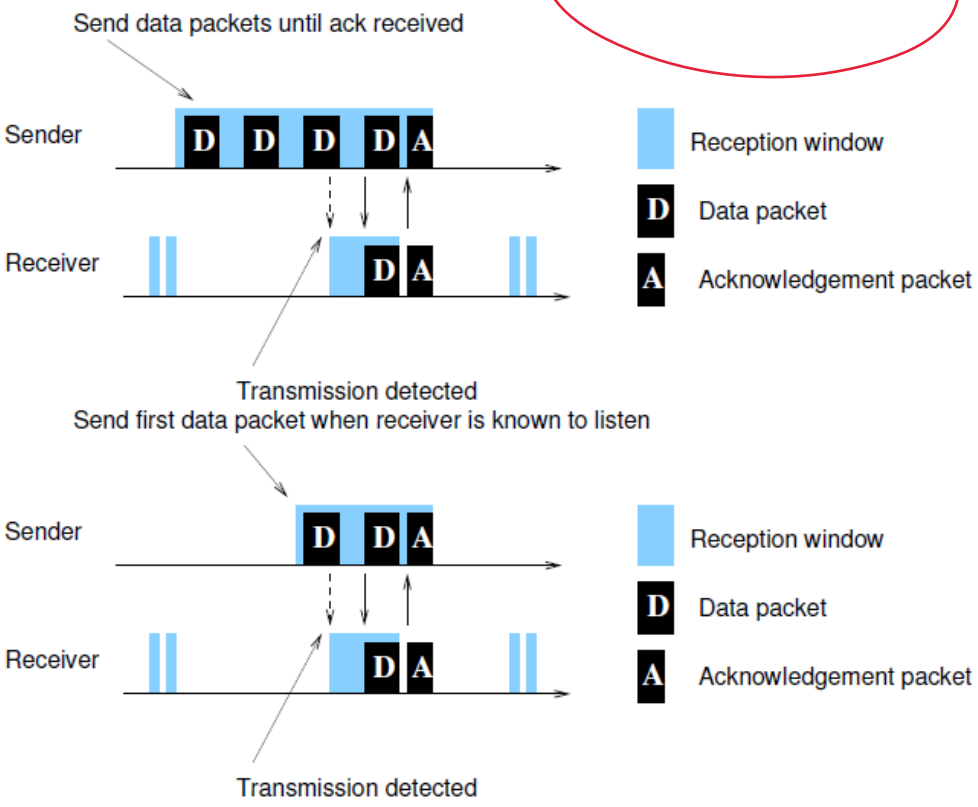


Figure 6: Transmission phase-lock: after a successful transmission, the sender has learned the wake-up phase of the receiver and subsequently needs to send fewer transmissions.